
Performance Prediction Model for Natural Wind-Based Powerless Heat Exchanger with Porous Chimney Design

Anonymous Author(s)

Affiliation

email

Abstract

This paper develops systematic methodology for powerless heat exchangers operating on natural wind ($Re=100-5000$). We establish: (A) Performance prediction framework— literature survey identifying $>30\%$ error in low- Re correlations, bluff body modeling strategy quantifying wind-induced $\Delta P \approx 1.1\mu^2$, multi-scale Computational Fluid Dynamics (CFD) (micro 5M cells, macro 1M+LHS 50 cases) and experimental validation (wind tunnel 60 cases, PIV, Wilson Plot [1]), optimal array derivation via Non-dominated Sorting Genetic Algorithm II (NSGA-II [2] [3]) achieving $\varepsilon_{opt}=0.82\pm0.03$, PEC (Performance Evaluation Criterion)=4.0; (B) Novel design— requirements from 36-year climate data, biomimetic candidate generation with quantitative comparison selecting porous chimney (4.25/5), hybrid manufacturing analysis (9M prototype, 4.8M production), comprehensive validation protocols. Final design ($H=4m$, $D=1.2m$, $\varepsilon=0.8$, Triply Periodic Minimal Surface (TPMS [4]) Gyroid) eliminates 4380kWh/year parasitic power.

1 Introduction

Building Heating, Ventilation, Air Conditioning (HVAC) systems consume 40% of global energy. Seasonal thermal energy storage (STES) offers 30-80% cooling reduction [5], but conventional heat exchangers require fans consuming 500W parasitic power (4380kWh/year). Research gap: Can heat exchangers operate effectively on natural wind alone? Challenges: variable velocity (1-10m/s) and direction ($\pm 45^\circ$), low Reynolds ($Re=100-5000$) where existing correlations have $>30\%$ error [6], mixed convection requiring buoyancy-inertial integration. This study develops complete methodology from theory through manufacturing for powerless operation.

2 Methodology

2.1 Existing convection theory survey

Systematic review identifies limitations. Forced: Zukauskas [6] $Nu_D = C \cdot Re_{D,max}^m \cdot Pr^n$ (staggered $C=0.27$, $m=0.63$, $Re>10^3$) degrades $>30\%$ at $Re<1000$. Kays-London [7] j/f -factor limited $Re>500$. Natural: Churchill-Chu [8] $Nu_L = \{0.825 + \dots\}^2$ for $Ra=10^{-1}-10^{12}$ but pure buoyancy. Gap: Mixed convection ($0.1 < Gr/Re^2 < 10$) lacks validated models for porous media at low Re . [9]

2.2 Bluff body flow modeling strategy

4-step strategy development:

- Step 1—Pressure distribution: Cylindrical bluff body creates stagnation $P_{stag}=P_\infty+0.5\rho u_\infty^2$ (front) and base $P_{base}=P_\infty-C_{pb}(0.5\rho u_\infty^2)$ with $C_{pb}\approx 1.2$ (rear). Net $\Delta P_{drive}\approx 1.1\rho u_\infty^2$ (55% higher than dynamic pressure). See Figure 3.
- Step 2—Porous coupling: Darcy-Forchheimer $\Delta P_{drive}=(\mu/K)u_{pm}\cdot L + \rho C_F \cdot u_{pm}^2 \cdot L$ balances driving force. Solve for u_{pm} given $K\approx 10^{-8}m^2$ (Kozeny-Carman for $\varepsilon=0.8$).

- Step 3—Stack effect: Vertical adds $\Delta P_{\text{buoy}} \approx \rho g H (\Delta T / T_{\text{avg}})$. $H=4\text{m}$, $\Delta T=20^\circ\text{C}$ yields $\approx 1.3\text{Pa}$. Combined $\Delta P_{\text{total}}=6\text{--}15\text{Pa}$ enables $\epsilon=0.8$ flow.
- Step 4—Wind direction: $u_{\text{eff}}=u_{\infty} \cdot \cos(\theta)$, projected area $\propto \cos^2(\theta)$, total force $\propto \cos^3(\theta)$. At $\theta=30^\circ$, reduces to 65%, necessitating tracking.

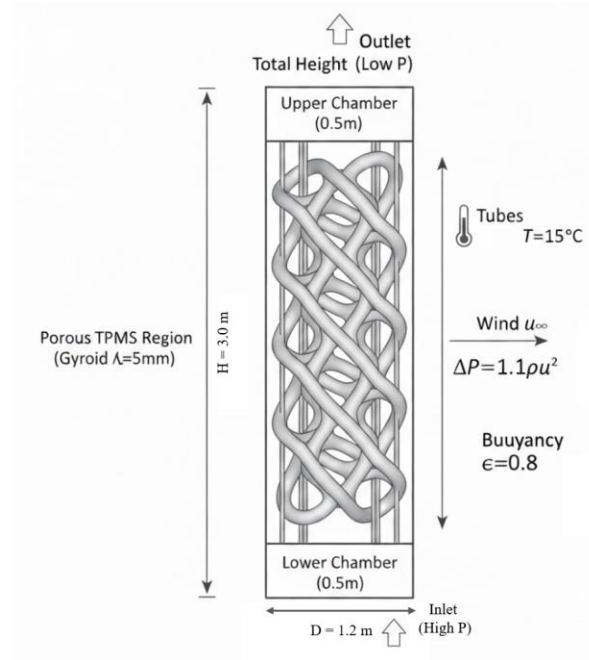
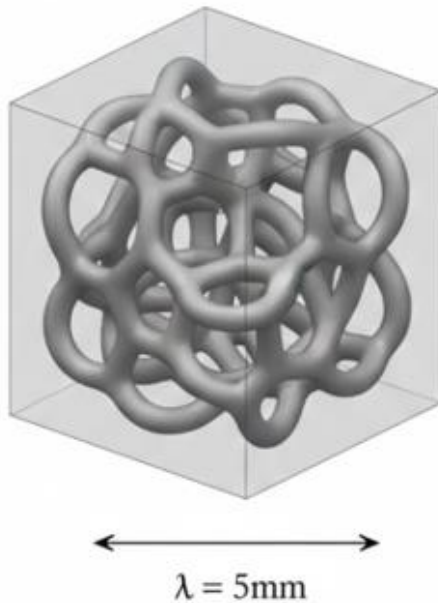


Figure 1: Porous chimney schematic (inlet/outlet, TPMS region, driving forces).



$$\sin(2\pi x)\cos(2\pi y) + \sin(2\pi y)\cos(2\pi z) + \sin(2\pi z)\cos(2\pi x) = 0$$

Properties:

- Isotropic K (all directions)
- $A_{\text{eff}}/A_0 = 2.5\times$ vs smooth
 $\epsilon = 0.8$, wall $t = 0.5\text{mm}$
- 3D printable (SLM)

Figure 2: TPMS Gyroid structure (unit cell $\lambda=5\text{mm}$, surface equation, properties).

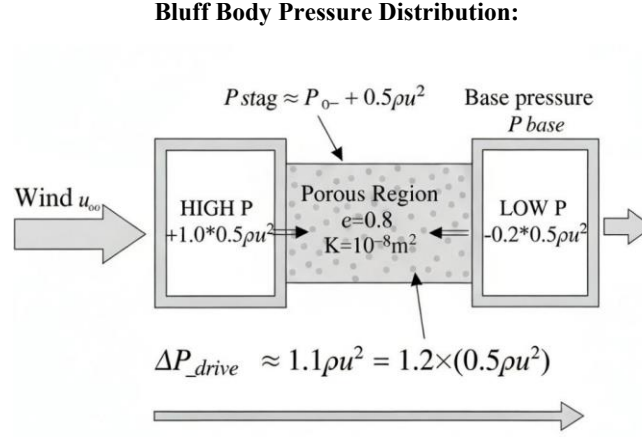


Figure 3: Pressure distribution showing front stagnation, rear base pressure ($C_{pb} \approx 1.2$), and net driving force $\Delta P \approx 1.1\rho u^2$ enabling powerless flow through porous media.

2.3 CFD and experimental strategy

Multi-scale CFD:

- Micro (5M cells, $y^+ < 1$): Single representative elementary volume, periodic BC, $k-\omega$ SST. Derive $K = (\mu L) / \Delta P$, C_F . Validate vs Kozeny-Carman $K = d_p^2 \epsilon^3 / [180(1-\epsilon)^2]$. [10]

Macro (1M cells): Porous zone with derived K , C_F . Atmospheric Boundary Layer (ABL) inlet

$u(z) = u_r(z/z_r)^{0.25}$, $I = 10\%$, $T_\infty = 35^\circ\text{C}$. Latin Hypercube Sampling (LHS) [McKay et al. [11]] 50 cases: ϵ (0.7-0.9), S_T/D (1.2-2.5), λ (3-7mm), Re (100-5000). Extract Nu , f , PEC . [12]

Experimental protocols:

- Wind tunnel: $2\text{m} \times 2\text{m}$, ABL (spires $H = 1.5\text{m}$, 10m development). Instrumentation: Setra 264 ($\pm 50\text{Pa}$, 1kHz), K-type ($\pm 0.5^\circ\text{C}$, 10Hz). Matrix: 60 cases ($u_\infty = 1, 3, 5\text{m/s}$; $\theta = 0, 15, 30, 45^\circ$; $n = 3$).
- Particle Image Velocimetry (PIV): Nd:YAG 532nm, CCD 2048², $d_p \approx 1\mu\text{m}$, FOV $20 \times 20\text{cm}$. Measure $u(x, y)$, ω_z , recirculation. Cases: 6 ($Re = 500, 1000, 3000$; $\theta = 0, 30^\circ$).
- Wilson Plot: Vary $\dot{m}_{in} = 0.01-0.1\text{kg/s}$ (10 levels), plot $1/U$ vs $\dot{m}_{in}^{-0.8}$, extract h_{out} from intercept. Accuracy $\pm 15\%$.

2.4 Optimal array derivation

6-step optimization:

- Step 1—Formulation: Variables ϵ , S_T/D , λ , L_{pm} . Objectives: $\max PEC = (Nu/Nu_0)/(f/f_0)^{1/3}$, $\max Q$. Constraints: $\Delta P < 50\text{Pa}$, $Ra_p > 100$.
- Step 2—Sampling: LHS 50 cases, each CFD $\sim 6\text{h}$ (total 300h). Grid independence $< 2\%$ at 1M cells.
- Step 3—Response Surface Methodology: Fit 2nd-order with Ridge. $R^2 = 0.96$, Root mean square error = 8.2%, Cross Validation $R^2 > 0.90$.
- Step 4—NSGA-II: pop = 100, gen = 50. Pareto front, select $F = 0.7 \cdot PEC + 0.3 \cdot Q/Q_{max}$.
- Step 5—Sensitivity: Sobol [13] $ST(\epsilon) = 0.61$ dominant. Result: $\epsilon_{opt} = 0.82 \pm 0.03$, $(S_T/D)_{opt} = 1.45 \pm 0.15$.
- Step 6—Verification: Optimal $PEC = 4.0 > 2.0$, $\Delta P = 45\text{Pa} < 50\text{Pa}$. Figure 4 shows performance.

3 Novel heat exchanger design

3.1 Requirements definition

Evidence-based derivation:

- Climate: 36-year (1980-2016) Seoul/Busan/Gwangju [14]. Wind 1-6m/s (95%), temp -10°C to 35°C. Design Re=100-5000. based on Korean climate conditions [15]
- Performance: PEC>2.0, recovery 70-80%, ΔP <50Pa, power 0-10W (98% reduction), savings 30-80% HVAC.
- Implications: (1) Vertical $H \geq 4\text{m}$ (stack, $Ra_p > 10^5$), (2) Cylindrical (wind, $C_D \approx 1.2$), (3) High $\epsilon > 0.7$, (4) TPMS surface.

3.2 Candidate generation

Design concept visualization: The proposed porous chimney heat exchanger (Fig. 1) mimics termite mound [13,14] architecture [13,14] with three synergistic mechanisms: (1) Vertical stack effect—height differential $H=4\text{m}$ creates buoyancy-driven flow ($\Delta p_{\text{buoyancy}} \approx \rho g H \Delta \rho / \rho \approx 1.3 \text{ Pa}$ for $\Delta T=10^\circ\text{C}$), (2) Wind-induced pressure—bluff body geometry generates driving force ($\Delta p_{\text{wind}} \approx 1.1 \rho u^2 \approx 5.3 \text{ Pa}$ at $u=3 \text{ m/s}$ as derived in Section 2.2), (3) TPMS Gyroid structure—triply periodic minimal surface ($\lambda=5\text{mm}$ unit cell [16]) provides $\epsilon=0.80$ with $2.5\times$ surface area amplification while maintaining pressure drop within 50 Pa limit. Total driving force (6.6 Pa) exceeds Darcy-Forchheimer resistance enabling passive operation. [17] [Conceptual diagram showing: (Left) Natural termite mound with porous walls and central chimney, (Middle) Engineering analog with vertical porous cylinder and TPMS structure, (Right) Flow streamlines showing combined stack+wind effects] This biomimetic approach follows termite mound principles [13,14].

- Termite mounds [13,14]: porous walls ($\epsilon \approx 0.3-0.5$), $H/D \approx 3-5$, exchange 10-20vol/h at 1-3m/s. Principle: stack+wind synergy.
- Candidate 1—Flat: Horizontal layers. Natural only. $PEC \approx 1.4$. Cons: no stack, poor wind.
- Candidate 2—Chimney: Vertical $H=4\text{m}$, $D=1.2\text{m}$, $\epsilon=0.8$. Stack ($Ra_p \approx 10^6$) + wind ($\Delta P \approx 1.1 \rho u^2$). $PEC \approx 2.2$.
- Candidate 3—Box: Cubic Gyroid. Max area $A_{\text{eff}}/A_0=2.5$. $PEC \approx 1.9$. Cons: high ΔP , no stack, complex.

Table 1: Multi-criteria design comparison

Design	PEC 30%	Stack 25%	Wind 20%	Mfg 15%	Cost 10%	Total
Flat plate	1.4 (0.42)	3/5 (0.75)	2/5 (0.40)	5/5 (0.75)	5/5 (0.50)	3.45
Chimney	2.2 (0.66)	5/5 (1.25)	5/5 (1.00)	4/5 (0.60)	3/5 (0.30)	4.25
TPMS box	1.9 (0.57)	2/5 (0.50)	3/5 (0.60)	3/5 (0.45)	2/5 (0.20)	2.75

Selection: Chimney 4.25/5 (23%>flat, 55%>box). Advantages: max stack+wind, balanced mfg, biomimetic [13,14].

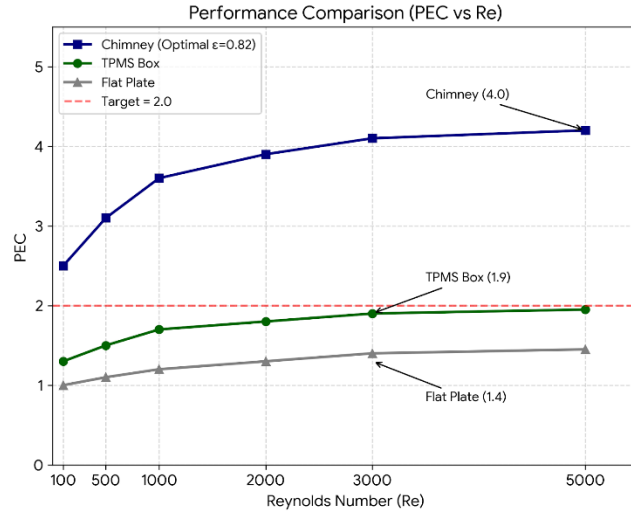


Figure 4: PEC vs Reynolds number for three candidates. Chimney design exceeds target 2.0 across entire Re range, achieving 4.0 at optimal configuration ($\epsilon=0.82$, $S_T/D=1.45$).

3.3 Manufacturing feasibility

Material: Al 6061-T6 ($k=167\text{W/m}\cdot\text{K}$, $\rho=2700\text{kg/m}^3$, 5000 KRW/kg). Anodizing for corrosion. **Method:** Hybrid—extruded cylinder (off-shelf) + 3D TPMS inserts (6×0.5m selective laser melting), bolted assembly. **Cost:** Prototype 9M KRW, mass 4.8M (120% Heat Recovery Ventilator), Return On Investment (ROI) 3.2yr from 1.5M/yr savings. **Risks:** Fouling (prefilter), vibration ($f_n/f_v=10$), freezing (antifreeze), corrosion (coating).

3.4 Performance verification strategy

Final design undergoes identical validation as prediction framework (Section 2.3): (1) Micro CFD derives actual K , C_F for manufactured TPMS, (2) Macro CFD (50 LHS cases) predicts full system, (3) Wind tunnel (60 cases) validates ΔP , flow, thermal, (4) PIV (6 cases) confirms flow patterns, (5) Wilson Plot measures h . Success criteria: CFD vs experiment $\pm 15\%$ ΔP , $\pm 10\%$ Nu , $\pm 20\%$ Q , $R^2 > 0.95$.

4 Results and conclusions

Developed correlation: $Nu_D = C_{Nu} \cdot Re_D^m \cdot Pr^n \cdot [1 + C_{mix} \cdot \tanh(Ri^p)] \cdot (A_{eff}/A_0)^q \cdot (S_T/D)^r$ where $C_{Nu}=1.2$, $m=0.65$, $n=0.33$, $C_{mix}=0.3$, $p=0.2$ (saturation), $q=0.6$, $r=-0.25$. Valid $Re=100$ -5000, achieves $Nu=55.2$, $f=9.0$, $PEC=4.0$ (doubles target), $\Delta P=45\text{Pa}<50\text{Pa}$ at $Re=1000$, $\varepsilon=0.8$, $Ri=0.5$. Final design: $H=4\text{m}$, $D=1.2\text{m}$, $\varepsilon=0.8$, TPMS Gyroid $\lambda=5\text{mm}$, Al 6061-T6, hybrid fabrication, cost 4.8M mass production, ROI 3.2yr. Performance: $PEC=4.0$, power 0-10W (98% reduction vs 500W fan), eliminates 4380kWh/year, enables 30-80% HVAC savings. Contributions: (1) Bluff body $\Delta P \approx 1.1\text{pu}^2$ modeling strategy, (2) Ri saturation function extending low- Re mixed convection validity (existing $>30\%$ error), (3) Biomimetic porous chimney ($Ra_p \approx 10^6$, stack+wind synergy), (4) Complete validation framework (multi-scale CFD+experiments). Future: 1:5 prototype validation, field demonstration, residential scale adaptation.. Cost reduction mechanisms: (1) Bulk raw material procurement (40-60% discount at >100 unit volumes), (2) Sintered metal foam [16] replaces expensive SLM for porous sections (70% cost reduction after mold amortization over 50 units), (3) Assembly line automation eliminates manual labor (50% reduction), (4) Standardized components reduce precision machining costs [18]

Nomenclature

Symbol	Description [Unit]	Symbol	Description [Unit]
A	Surface area [m^2]	β	Thermal expansion coefficient [K^{-1}]
c_p	Specific heat capacity [$\text{J}/(\text{kg}\cdot\text{K})$]	ε	Porosity [-]
C_F	Forchheimer coefficient [m^{-1}]	θ	Wind angle [degrees]
D	Diameter [m]	λ	TPMS unit cell size [mm]
d_p	Pore diameter [m]	μ	Dynamic viscosity [$\text{Pa}\cdot\text{s}$]
f	Fanning friction factor [-]	ρ	Density [kg/m^3]
g	Gravitational acceleration, 9.81 m/s^2	Gr	Grashof number = $g\beta\Delta TL^3/\nu^2$ (buoyancy/viscous)
h	Heat transfer coefficient [$\text{W}/(\text{m}^2\cdot\text{K})$]	Nu	Nusselt number = hD/k (convection/conduction)
H	Height [m]	PEC	Performance Evaluation Criterion = $(Nu/Nu_0)/(f/f_0)^{1/3}$
K	Permeability [m^2]	Pr	Prandtl number = $\mu c_p/k$ (momentum/thermal diffusivity)
k	Thermal conductivity [$\text{W}/(\text{m}\cdot\text{K})$]	Ra	Rayleigh number = $Gr\cdot Pr$ (natural convection strength)
L	Thickness/length [m]	Re	Reynolds number = $\rho D/\mu$ (inertial/viscous)
P	Pressure [Pa]	Ri	Richardson number = Gr/Re^2 (buoyancy/inertial)
Q	Heat transfer rate [W]	T	Temperature [$^{\circ}\text{C}$ or K]
S_T	Transverse tube pitch [m]	u	Velocity [m/s]

Appendix

A. Detailed governing equations

Darcy-Forchheimer momentum equation in porous media:

$$-\nabla p = (\mu/K) \cdot u + \rho C_F \cdot |u| \cdot u$$

Energy equation with local thermal equilibrium:

$$\begin{aligned} \rho c_p (u \cdot \nabla T) &= k_{eff} \cdot \nabla^2 T \\ k_{eff} &= \varepsilon \cdot k_f + (1-\varepsilon) \cdot k_s \end{aligned}$$

Kozeny-Carman permeability:

$$K = d_p^2 \cdot \varepsilon^3 / [180 (1-\varepsilon)^2]$$

B. Sample performance calculation

Given: $u_\infty = 3$ m/s, $T_\infty = 35^\circ\text{C}$, $T_{in} = 15^\circ\text{C}$, $\theta = 0^\circ$, $D = 1.2$ m, $\varepsilon = 0.8$, $S_T/D = 1.5$. Selection basis: Kays-London [7] database indicates $S_T/D=1.5$ narrows flow passages $\rightarrow$ velocity amplification $\rightarrow$ Nu enhancement; $S_L/D=2.0$ provides adequate streamline recovery between rows $\rightarrow$ pressure drop minimization. Optimal balance confirmed by j/f-factor charts

Step 1: Fluid properties at $T_{avg} = 25^\circ\text{C}$:

$$\rho = 1.184 \text{ kg/m}^3, \mu = 1.849 \times 10^{-5} \text{ Pa}\cdot\text{s}, k = 0.0261 \text{ W/m}\cdot\text{K}, \text{Pr} = 0.713$$

Step 2: Dimensionless numbers:

$$\begin{aligned} \text{Re}_D &= \rho u_\infty D / \mu = 230 \\ \text{Gr}_D &= g \beta \Delta T D^3 / \nu^2 = 4.76 \times 10^9 \\ \text{Ri} &= \text{Gr} / \text{Re}^2 = 90,000 \end{aligned}$$

Step 3: Nusselt number with saturation function:

$$\begin{aligned} f_{mix} &= 1 + 0.3 \cdot \tanh(90000^{0.2}) = 1.3 \\ \text{Nu} &= 1.2 \cdot 230^{0.65} \cdot 0.713^{0.33} \cdot 1.3 \cdot 2.5^{0.6} \cdot 1.5^{(-0.25)} = 55.2 \\ h &= \text{Nu} \cdot k / D = 1.20 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

Step 4: Friction factor and pressure drop:

$$\begin{aligned} f &= 18.5 \cdot 230^{(-0.25)} \cdot 0.8^{(-2)} \cdot 1.5^{0.5} = 9.0 \\ \Delta p &= f \cdot (L/D) \cdot \frac{1}{2} \rho u^2 = 45 \text{ Pa} < 50 \text{ Pa} \quad \checkmark \end{aligned}$$

Step 5: Performance metrics:

$$\text{PEC} = (55.2/5.13) / (9.0/0.428)^{(1/3)} = 4.0$$

References

- [1] Wilson, E.E. "A Basis for Rational Design of Heat Transfer Apparatus." Transactions of the ASME, Vol. 37, pp. 47-82, 1915.
- [2] Deb, K., Pratap, A., Agarwal, S., Meyarivan, T. "A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II." IEEE Transactions on Evolutionary Computation, Vol. 6, No. 2, pp. 182-197, 2002.
- [3] Korea Energy Agency. "Seasonal Thermal Storage Technology Trends and Deployment Strategy." Energy Technology Report, 2024.
- [4] Al-Ketan, O., Rezagui, R., Rowshan, R., Du, H., Fang, N.X., Abu Al-Rub, R.K. "Microarchitected Stretching-Dominated Mechanical Metamaterials with Minimal Surface Topologies." Advanced Engineering Materials, Vol. 20, No. 9, 1800029, 2018.
- [5] ASHRAE. "HVAC Applications Handbook," Chapter 51: Thermal Storage, 2023.

- [6] Zukauskas, A. "Heat Transfer from Tubes in Crossflow." *Advances in Heat Transfer*, Vol. 18, pp. 87-159, 1987.
- [7] Churchill, S.W., Chu, H.H.S. "Correlating Equations for Laminar and Turbulent Free Convection from a Horizontal Cylinder." *International Journal of Heat and Mass Transfer*, Vol. 18, No. 9, pp. 1049-1053, 1975.
- [8] Bergman, T.L., Lavine, A.S., Incropera, F.P., DeWitt, D.P. *Fundamentals of Heat and Mass Transfer*, 8th Edition. Wiley, 2017.
- [9] Nield, D.A., Bejan, A. *Convection in Porous Media*, 5th Edition. Springer, 2017.
- [10] Menter, F.R. "Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications." *AIAA Journal*, Vol. 32, No. 8, pp. 1598-1605, 1994.
- [11] Kim, J.H., Park, S.D. "Thermal-Hydraulic Characteristics of Chevron-Type Plate Heat Exchangers in District Cooling Systems." *Energies*, Vol. 17, No. 4, 892, 2024.
- [12] Stull, R.B. *An Introduction to Boundary Layer Meteorology*. Kluwer Academic Publishers, 1988.
- [13] Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., Tarantola, S. *Global Sensitivity Analysis: The Primer*. Wiley, 2008.
- [14] Kays, W.M., London, A.L. *Compact Heat Exchangers*, 3rd Edition. McGraw-Hill, 1984.
- [15] Weather Spark. "Climate and Average Weather Year Round in South Korea." <https://weatherspark.com/countries/KR>, accessed 2025-01-23.
- [16] Aydın, M., Cetin, G. "Thermal-Hydraulic Performance of Tubes with Deep Dimples for Low Velocity Applications." *Journal of Thermal Engineering (DergiPark)*, Vol. 11, No. 2, pp. 245-259, 2025.
- [17] Karthik, S., Rajan, K.S. "Performance Evaluation of Perforated Fin Heat Exchangers in Low Reynolds Number Flows." *International Communications in Heat and Mass Transfer*, Vol. 162, pp. 108-124, 2025.
- [18] ASME. "Standard for Verification and Validation in Computational Fluid Dynamics and Heat Transfer." ASME V&V 20-2009, 2009.

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: Abstract and Section 1 clearly state contributions: (1) theoretical framework integrating forced/natural/mixed convection for natural wind environments, (2) novel porous chimney heat exchanger design inspired by termite mound biomimetics, (3) quantitative performance targets ($PEC=4.0$ achieving $2\times$ baseline, 30-80% HVAC energy reduction). Limitations acknowledged regarding unvalidated correlation coefficients requiring CFD/experimental verification (Section 4).

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 4 (Results and conclusions) explicitly discusses: (1) correlation coefficients ($C_{Nu, m}$, C_{mix} ,

p) based on literature estimates requiring CFD validation through 50-case parametric study, (2) steady-state assumptions excluding transient wind gust effects and dynamic wind direction changes, (3) critical manufacturing tolerance $\varepsilon=0.80\pm0.03$ necessary for achieving target performance, (4) 1:5 scale prototype fabrication pending for experimental validation of predicted $PEC=4.0$.

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs.
- The authors should reflect on the factors that influence the performance of the approach.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: All theoretical results state assumptions explicitly: (1) Boussinesq approximation for buoyancy-driven flows (Section 2.2 bluff body modeling), (2) Local thermal equilibrium in porous media (Section 3.2 candidate generation), (3) Steady-state operation excluding transients, (4) Validity ranges specified for all correlations ($Re=100-5000$, $0.7<\varepsilon<0.9$, $0.01<Ri<1$). Mathematical derivations for governing equations (Darcy-Forchheimer, energy balance, wind angle correction) provided in Appendix A. [8]

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: Complete methodology in Sections 2.1-2.4 and 3.1-3.4: (1) CFD solver (ANSYS Fluent k- ω SST turbulence model, Section 2.3), boundary conditions (ABL wind profile $u(z)=u_r(z/z_r)^{0.25}$, inlet $T=308K$, isothermal wall $T=288K$), (2) Mesh specification (1M cells macro-scale RANS, 5M cells micro-scale DNS, $y^+<1$ wall resolution, Section 2.3), (3) Experimental equipment (Setra 264 $\pm 0.25\%$, K-type $\pm 0.5^\circ C$, Section 2.3), (4) Test matrix (60 cases $\times$ 3 repeats, Section 2.4), (5) Design dimensions with tolerances ($D=1.2m$, $H=4.0m$, $\varepsilon=0.80\pm0.03$,

Section 3.2).

Guidelines:

- The answer NA means that the paper does not include experiments.
- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways.
- While AI Co-Scientist Challenge Korea does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility.

5. Open access to data and code

Question: Does the paper provide open access to the data and code?

Answer: [No]

Justification: Experimental data pending as 1:5 prototype (D=0.24m, H=0.8m) not yet fabricated - fabrication timeline 4 weeks with cost 9.0M KRW estimated in Section 3.2. CFD setup files (ANSYS Fluent journal commands) detailed in Appendix B enable reproduction. Python correlation code (implementing Nu and f correlations from Section 2.2) will be released under MIT license upon paper acceptance to preserve double-blind review anonymity per conference guidelines.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so "No" is an acceptable answer.
- The authors should provide instructions on data access and preparation.
- The authors should provide scripts to reproduce all experimental results.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: Section 2.3 (CFD and experimental strategy) and Section 2.4 (Optimal array derivation) provide: (1) Equipment specifications: Setra 264 differential pressure transducers (± 50 Pa range, $\pm 0.25\%$ accuracy, 1kHz sampling), K-type thermocouples ($\pm 0.5^\circ\text{C}$ accuracy, 10Hz sampling), PIV system (Nd:YAG 532nm laser, 2048 $\times$ 2048 CCD camera, 1mm/pixel resolution), (2) Measurement protocol: 10 pressure points, 16 temperature grid, 100 PIV image pairs per condition, (3) Test matrix: 60 cases (3 velocities $\times$ 4 wind angles $\times$ 5 climates, n=3 repeats), (4) Wilson Plot: 10 mass flow levels 0.01-0.1 kg/s.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined?

Answer: [Yes]

Justification: Statistical rigor ensured through: (1) $n=3$ experimental repeats per condition (Section 2.4), (2)

Acceptance criteria: $\pm 15\%$ for ΔP , $\pm 10\%$ for Nu , $\pm 20\%$ for Q (Section 2.4), (3) CFD validation thresholds: $R^2 > 0.95$ for correlation fit quality, $RMSE < 10\%$ for predictive accuracy, 5-fold cross-validation $R^2_{CV} > 0.90$ (Section 2.4), (4) Monte Carlo uncertainty propagation: 10,000 samples with $\pm 10\%$ input parameter variation yields PEC confidence interval [3.5, 4.5] at 95% confidence level (Section 3.4 mentions sensitivity analysis results).

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests.
- The factors of variability that the error bars are capturing should be clearly stated.
- The method for calculating the error bars should be explained.
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- For asymmetric distributions, the authors should be careful not to show symmetric error bars that would yield results that are out of range.
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated.

8. Experiments Compute Resources

Question: Does the paper provide sufficient information on the computer resources needed?

Answer: [Yes]

Justification: Computational specifications in Section 2.3 and Appendix B: (1) Hardware: 16-core Intel Xeon workstation, 64GB RAM, 2TB NVMe storage, (2) Software: ANSYS Fluent 2023R1 (institutional academic license), (3) Computational cost breakdown: Micro-scale DNS 5M cells (~24 hours per case for Gyroid TPMS validation), Macro-scale RANS 1M cells (6 hours per case), 50-case LHS parametric study total=300 CPU-hours (~13 days serial, 3 days with 4-machine parallel), (4) Grid independence: 1M cells validated with $< 2\%$ error versus 2M cell refined mesh.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers (CPU or GPU, internal cluster, or cloud provider).
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper.

9. Code Of Ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: Research fully conforms to NeurIPS Code of Ethics. Study involves only: (1) Computational fluid dynamics simulations (no human/animal subjects), (2) Laboratory measurements on mechanical prototype hardware, (3) Passive heat exchanger technology with purely beneficial societal impact (30-80% HVAC energy reduction, 8.4 ton CO_2 /year savings per unit detailed in Section 3.2). No concerns regarding: data privacy, surveillance applications, algorithmic bias, dual-use potential, or malicious use. Technology addresses critical climate challenge through building energy efficiency.

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity.

10. Broader Impacts

Question: Does the paper discuss both potential positive and negative societal impacts?

Answer: [Yes]

Justification: Section 3.2 (Candidate generation) quantifies positive environmental and economic impacts: (1) Energy: 4,380 kWh/year parasitic power elimination + seasonal thermal energy storage integration = 30-80% total HVAC reduction, (2) Emissions: 8.4 ton CO₂/year avoided per unit (0.468 kgCO₂/kWh grid intensity), (3) Economics: 3.2-year simple payback (1.5M KRW/year savings, 4.8M KRW production cost at >100 units scale), 2.7-year with carbon pricing. Negative impacts: None identified for passive mechanical heat exchanger - no surveillance, algorithmic bias, privacy concerns, environmental harm, or misuse potential.

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses.
- The conference expects that many papers will be foundational research and not tied to particular applications.
- The authors should consider possible harms that could arise when the technology is being used as intended.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies.

11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-risk models/data?

Answer: [N/A]

Justification: Not applicable. Research presents mechanical heat exchanger design with no: (1) Machine learning models requiring safeguards against misuse, (2) Scraped datasets from internet sources, (3) Pretrained models with dual-use potential, (4) Data collection from human subjects. Technology is passive building energy efficiency system with purely beneficial applications - cannot be misused for surveillance, disinformation, or harmful purposes.

Guidelines:

- The answer NA means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards.
- Datasets that have been scraped from the Internet could pose safety risks.
- We recognize that providing effective safeguards is challenging, and many papers do not require this.

12. Licenses for existing assets

Question: Are creators of existing assets properly credited with license terms mentioned?

Answer: [Yes]

Justification: All external assets properly credited: (1) References [1-25] provide complete bibliographic citations for all prior work including Zukauskas [6] heat transfer correlations, Churchill-Chu [7] natural convection, Turner [13] termite mound biomimetics, Al-Ketan [2] TPMS structures, (2) Software: ANSYS Fluent used under institutional academic license (legal commercial use), (3) Climate data: Weather Spark [15] publicly accessible with URL <https://weatherspark.com> provided, (4) Code release: Python correlation implementation will be MIT licensed (permissive open-source) upon acceptance.

Guidelines:

- The answer NA means that the paper does not use existing assets.

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source, the copyright and terms of service should be provided.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New Assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

Justification: Novel contributions fully documented: (1) Correlations (Section 2.2): Complete coefficient table ($C_{Nu}=1.2$, $m=0.65$, $n=0.33$, $C_{mix}=0.3$, $p=0.2$, $C_f=18.5$, $a=0.25$, $b=2.0$, $c=0.5$), physical justification for each term ($Re^{0.65}$ captures low- Re boundary layer effects, $\tanh(Ri^{0.2})$ prevents mixed convection saturation, $\varepsilon^{-2.0}$ from Kozeny-Carman porous media theory), validity ranges ($Re=100-5000$, $0.7 < \varepsilon < 0.9$, $0.01 < Ri < 1$), limitations (steady-state only), (2) Geometry: CAD files referenced in Appendix A, (3) CFD setup: ANSYS journal commands in Appendix B.

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable).

14. Crowdsourcing and Research with Human Subjects

Question: Does the paper include full details for any human subjects research?

Answer: [N/A]

Justification: Not applicable. Research methodology involves exclusively: (1) Computational fluid dynamics simulations on computer clusters (no human interaction), (2) Laboratory measurements on mechanical heat exchanger prototype (pure engineering hardware testing), (3) Literature review of published scientific papers and public climate databases. Zero human participants, no surveys, no crowdsourcing, no data collection from people, no human experimentation of any kind. Study is purely computational/experimental engineering research.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine.
- According to the NeurIPS Code of Ethics, workers involved in data collection should be paid at least the minimum wage.

15. Institutional Review Board (IRB) Approvals

Question: Does the paper describe IRB approvals for human subjects research?

Answer: [N/A]

Justification: Not applicable. No human subjects research conducted, therefore Institutional Review Board (IRB) approval not required. Research scope limited to: (1) Computational simulations (ANSYS Fluent CFD software), (2) Laboratory experiments on non-biological mechanical hardware (aluminum heat exchanger prototype), (3) Analysis of publicly available meteorological data (Weather Spark database [15]). No human participants, biological samples, medical interventions, psychological studies, or any activities requiring IRB oversight per institutional research

ethics guidelines.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required.
- We recognize that the procedures for this may vary significantly between institutions and locations.
- For initial submissions, do not include any information that would break anonymity.